RUSLAN HASANOV

ML/DL Developer | Deep Learning, Computer Vision, Document AI

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SUMMARY

Machine learning engineer with hands-on deep learning experience across computer vision (CNN, YOLO, OCR) and applied ML pipelines, including a multi-modal AI system for extracting structured data from energy-sector drilling reports. Comfortable building, evaluating, and tuning models end-to-end in Python

SKILLS

Data Science & ML: Python, NumPy, pandas, scikit-learn, PyTorch, TensorFlow, Time Series Forecasting (LSTM, SARIMAX, Prophet), YOLO, OCR (PaddleOCR, EasyOCR, PyMuPDF), Image Classification, Data Wrangling & Validation
Statistics & Math: Regression Analysis, Statistical Distributions, Hypothesis Testing, Linear Algebra Fundamentals
Data Visualization: Matplotlib, Seaborn
Databases & Tools: SQL, Git, Docker

PROFESSIONAL EXPERIENCE

Junior Python Developer, EPAM| Baku, Azerbaijan **04/2026 - 09/2026**

- Built FastAPI endpoints for restaurant listings, reservations, and booking rules
- Migrated backend from AWS Lambda/DynamoDB to Kubernetes, Docker, MongoDB
- Secured APIs with JWT, added RabbitMQ messaging, wrote pytest test coverage
- Designed and built an end-to-end AI analysis pipeline on EPAM DIAL, covering prompt design, LLM API integration, and structured output parsing for downstream consumption by backend services

AI Engineer Trainee, Araz Supermarket MMC | Baku, Azerbaijan **10/2025 - 01/2026**

- Built a GPT-based contract analyzer that flags legal risks and gaps from PDF/DOCX filings
- Developed an OCR pipeline (PyMuPDF, EasyOCR) extracting text from scanned PDFs, images, and DOCX CVs
- Built an LLM resume-screening tool scoring CVs against job descriptions and exporting ranked results

Research Data Analyst, S-MAT, University of Turku | Turku, Finland **01/2025 - 05/2025**

- Built a Tkinter desktop app to process J-V/P-V solar cell measurement data from Excel exports
- Automated computation of photovoltaic metrics (PCE, FF, Jsc, Voc) with J-V/P-V curve visualization
- Generated standardized PDF lab reports, cutting manual analysis work for a university research group

Data Science Trainee, Turku University of Applied Sciences l Turku, Finland **05/2024 - 08/2024**

- Collected and preprocessed household energy consumption data, engineering features for time series forecasting models
- Built LSTM, SARIMAX, and Prophet models for household energy forecasting
- Achieved sub-10% MAPE for next-day household energy consumption predictions

EDUCATION

Master of Science in Materials Engineering with a thematic module in Data Analytics **08/2023 - 08/2025**

University of Turku | https://www.utu.fi/en | GPA: 4.23 / 5

- Had coursework about Machine Learning, and AI in Renewable Energy. Acquired a minor in Data Analytics
- Thesis title: Data Analysis for Emerging Materials in Hybrid Perovskite Photovoltaics. (> 1500 downloads)

Bachelor of Engineering in Chemical Engineering **09/2018 - 06/2023**

Baku Higher Oil School | https://bhos.edu.az/en | GPA: 92 /100

- Thesis title: Design and Optimization of Toluene Reactor

RELEVANT PROJECTS

Hybrid Retrieval-Augmented Generation System for Document Q&A **09/2026**

- Built a hybrid RAG pipeline combining BM25 and vector search via Reciprocal Rank Fusion
- Added cross-encoder reranking, cutting false positives BM25 alone ranked incorrectly
- Achieved 1.0 faithfulness score (zero hallucination) via custom LLM-as-judge evaluation

AI System for Structuring & Querying Daily Drilling Reports (Multi-Modal Document AI) **02/2026**

- Built AI pipeline (YOLO + PaddleOCR + Gemini Vision) turning drilling report PDFs into SQL data
- Designed a TAG-style chatbot for natural-language querying with LLM-generated analytics summaries
- Delivered a Streamlit app with modular architecture for ingestion, domain modeling, and reporting

Satellite Image Classification with a ResNet18 model using PyTorch **08/2025**

- Trained a ResNet18 CNN to classify satellite imagery into 10 LULC classes, 98.9% test accuracy
- Built an evaluation pipeline with per-class accuracy, confusion matrices, and classification reports
- Achieved consistent F1-scores of 0.98-1.00 across all classes, showing strong generalization

Identification of Rice Varieties Using Machine Learning and Deep Learning Algorithms **03/2024**

- Engineered a CV pipeline extracting shape and color features to classify 5 rice grain varieties
- Applied PCA for dimensionality reduction to visualize and analyze class separability
- Trained and tuned KNN, Logistic Regression, and SVM classifiers via GridSearchCV

CERTIFICATES

Getting Started with Kubernetes **08/2026**

- Practised with Kubernetes cluster
- Built a microservices architecture
- Discovered how to secure Kubernetes clusters

Docker Foundations Professional Certificate **08/2026**

- Discovered how Docker is used in software development
- Practiced using containers in a real-world project
- Explored advanced deployment techniques with Docker Compose

AWS Knowledge: Serverless - Training Badge **05/2026**

- Developed technical skills and knowledge of AWS serverless concepts
- Learnt about services with a focus on AWS Lambda and Amazon API Gateway

Computer Vision Bootcamp, Turkish AI Community **02/2026**

- Completed intensive program on image processing and detection techniques
- Studied detection algorithms including YOLO for object recognition tasks

Databases for Developers, Oracle **11/2025**

- Completed training in relational database design and SQL (DDL, DML, joins, subqueries)

ADDITIONAL INFORMATION

Languages: Azerbaijani (Native), English (Near-Native, IELTS Academic 8.5 - 23/11/2024), Turkish (Fluent)

Hobbies and Personal Interests: Reading, Playing Guitar and interested in cultural exchange and volunteering activities